

## ENGINEERING CALCULATION NOTE

# Adjustable Motor Mount

## Preliminary Static Checks for the NEMA 17-Style Motor Mount

*Project calculation record | Status: provisional portfolio analysis*

### Scope and limitations

The 30 N lateral static load is a provisional portfolio design assumption.

The dry aluminum-to-aluminum friction coefficient of 0.20 is an engineering assumption and was not experimentally measured.

These results are simplified static checks intended to demonstrate basic design reasoning, not a certified structural analysis.

## Calculation basis

This note explains five calculations used to check the adjustment geometry and the simplified load path of the motor mount. Each section states the design question, its relevance, the assumptions, the governing equation, the numerical substitution, the result, and a short interpretation. No material strength allowables or experimentally measured preload values are introduced.

### 1. Adjustment travel

**What is being calculated:** The conservative usable linear movement provided by one adjustment slot.

**Why this is relevant to the design:** Adjustment travel determines how far the motor can be repositioned for alignment or belt-tension adjustment.

**Assumptions used:** The slot has a 24 mm overall length and 3 mm end radii. The distance between the two end-arc centers is taken as the usable travel.

**Governing equation:**  $s = L - 2r$

**Substitution with project values:**  $s = 24 \text{ mm} - 2(3 \text{ mm}) = 18 \text{ mm}$

**Final result:** Conservative usable linear adjustment travel: 18 mm

**Engineering interpretation:** The geometry provides 18 mm of center-to-center movement. Using this value avoids counting the rounded slot ends as additional usable travel.

## 2. Load distribution between the two M5 bolts

**What is being calculated:** The lateral force assigned to each of the two M5 adjustment bolts.

**Why this is relevant to the design:** The bearing-stress and bolt-shear checks require a force per bolt rather than the total applied load.

**Assumptions used:** The provisional lateral static load is 30 N. The two M5 bolts are assumed to share that load equally.

**Governing equation:**  $F_{\text{bolt}} = F_{\text{total}} / n$

**Substitution with project values:**  $F_{\text{bolt}} = 30 \text{ N} / 2 = 15 \text{ N}$

**Final result:** Load per M5 bolt: 15 N

**Engineering interpretation:** Each bolt is represented by a 15 N lateral load in the following simplified checks. Real assemblies may not share load perfectly, so this is an idealized static distribution.

## 3. Bearing stress at the Motor Plate slot

**What is being calculated:** The nominal bearing stress where one M5 bolt contacts the side of a Motor Plate slot.

**Why this is relevant to the design:** Bearing stress provides a simple measure of the local compressive stress transferred from the bolt into the slot wall.

**Assumptions used:** Force per bolt is 15 N, Motor Plate thickness is 5 mm, bolt diameter is 5 mm, and the projected bearing area is  $t d$ .

**Governing equation:**  $\sigma_b = F / (t d)$

**Substitution with project values:**  $\sigma_b = 15 \text{ N} / [(5 \text{ mm})(5 \text{ mm})] = 15 / 25 \text{ N/mm}^2$

**Final result:** Nominal bearing stress:  $0.60 \text{ N/mm}^2 = 0.60 \text{ MPa}$

**Engineering interpretation:** The calculated nominal bearing stress is 0.60 MPa for the approved provisional load. This is a simplified contact check and is not compared with a material allowable in this document.

## 4. Shear stress in each M5 bolt

**What is being calculated:** The average direct shear stress in the circular cross-section of each M5 bolt.

**Why this is relevant to the design:** This check expresses the 15 N per-bolt lateral load as an average shear stress in the bolt body.

**Assumptions used:** Each bolt carries 15 N, the bolt diameter is 5 mm, and the resisting area is the full nominal circular cross-section.

**Governing equation:**  $A = \pi d^2 / 4$ ;  $\tau = F / A$

**Substitution with project values:**  $A = \pi(5 \text{ mm})^2 / 4 = 19.63 \text{ mm}^2$ ;  $\tau = 15 \text{ N} / 19.63 \text{ mm}^2 = 0.764 \text{ N/mm}^2$

**Final result:** Average shear stress per M5 bolt: approximately 0.76 MPa

**Engineering interpretation:** The simplified average shear stress is approximately 0.76 MPa. No bolt-grade allowable or safety margin is calculated here because those checks were not included in the approved scope.

## 5. Friction and clamping force required to prevent sliding

**What is being calculated:** The idealized total clamping force required for friction to resist the 30 N lateral load, followed by the required share per bolt.

**Why this is relevant to the design:** The adjustable joint relies on friction between the clamped plate surfaces to hold the selected motor position.

**Assumptions used:** The lateral load is 30 N, the assumed dry aluminum-to-aluminum friction coefficient is 0.20, and two bolts provide equal clamping force. The coefficient was not experimentally measured.

**Governing equation:**  $F_{\text{friction}} = \mu N_{\text{total}}$ ;  $N_{\text{total}} = F_{\text{lateral}} / \mu$

**Substitution with project values:**  $N_{\text{total}} = 30 \text{ N} / 0.20 = 150 \text{ N}$ ;  $N_{\text{bolt}} = 150 \text{ N} / 2 = 75 \text{ N}$

**Final result:** Required total clamping force: 150 N; required clamping force per bolt: 75 N

**Engineering interpretation:** Under the assumed coefficient, a total normal clamping force of 150 N is required to resist sliding. The 75 N per-bolt value is a force requirement only; no tightening torque or achieved bolt preload is calculated.

## Results summary

| Check             | Result | Basis                                 |
|-------------------|--------|---------------------------------------|
| Adjustment travel | 18 mm  | 24 mm slot length and 3 mm end radius |
| Load per M5 bolt  | 15 N   | 30 N shared equally by two bolts      |

| Check                           | Result                     | Basis                                          |
|---------------------------------|----------------------------|------------------------------------------------|
| Motor Plate slot bearing stress | 0.60 MPa                   | 15 N over a 5 mm x 5 mm projected area         |
| M5 bolt shear stress            | approximately 0.76 MPa     | 15 N over a 19.63 mm <sup>2</sup> nominal area |
| Required clamping force         | 150 N total; 75 N per bolt | Dry friction coefficient assumed as 0.20       |

*These values support the basic portfolio design rationale only. They should be revisited if the real motor, operating load, surface condition, materials, or joint behavior differs from the approved assumptions.*